PROBLEM STATEMENT

Problem Statement 1 – Rain and Wet Ground

The objective of this problem is to implement the Resolution Algorithm for proving a conclusion in propositional logic. Consider a simple weather-related knowledge base containing two statements. First, there is a rule stating that if it rains, then the ground becomes wet. Second, it is given as a fact that it is raining.

The problem requires these statements to be represented using suitable propositional variables and converted into Conjunctive Normal Form (CNF). The goal is to prove that the ground is wet. To perform resolution, the negation of the goal is added to the knowledge base.

Problem Statement 2 – Student Assignment Submission

The objective of this problem is to implement the Resolution Algorithm to determine whether a conclusion about a student's internal marks logically follows from a given knowledge base. The knowledge base contains a rule stating that if a student submits an assignment, then the student receives internal marks. It is also given that Rahul has submitted the assignment.

The given statements must first be represented using propositional logic and converted into Conjunctive Normal Form (CNF). The goal is to prove that Rahul receives internal marks. Using the resolution-by-contradiction approach, the goal is negated and added to the set of clauses. The Resolution Algorithm is then applied by identifying complementary literals and generating new clauses.

Problem Statement 3 – Library Membership

The objective of this problem is to implement the Resolution Algorithm to prove a conclusion related to library membership. The knowledge base contains a rule stating that if a person is a library member, then the person can borrow books. It is also given as a fact that Priya is a library member.

The given statements must be represented using propositional logic and converted into Conjunctive Normal Form (CNF). The goal is to prove that Priya can borrow books. The negation of the goal is added to the knowledge base, and the Resolution Algorithm is applied by identifying complementary literals and generating new clauses.

If the resolution process produces the empty clause (□), it establishes a contradiction with the negated goal. Therefore, the original goal is logically proved. This problem demonstrates how logical reasoning can be applied to a simple library management scenario.

Problem Statement 4 – Placement Eligibility

The objective of this problem is to implement the Resolution Algorithm to prove a conclusion related to student placement eligibility. The knowledge base contains a rule stating that if a student clears the aptitude test, then the student is eligible for placement. It is also given as a fact that Arun has cleared the aptitude test.

The statements are first represented using propositional logic and converted into Conjunctive Normal Form (CNF). The goal is to prove that Arun is eligible for placement. To perform proof by contradiction, the goal is negated and added to the knowledge base. The Resolution Algorithm is then applied to the resulting clauses by resolving complementary literals.

When the resolution process derives the empty clause (□), it proves that the negation of the goal is inconsistent with the given facts and rules. Hence, the original conclusion that Arun is eligible for placement is logically established. This problem demonstrates the use of automated reasoning and logical inference in an academic placement scenario.

Problem Statement 5 – Access Control System

The objective of this problem is to implement the Resolution Algorithm for proving a conclusion in an access control system. The knowledge base contains two rules. The first rule states that if a user enters the correct password, the user is authenticated. The second rule states that if a user is authenticated, the user is granted access. It is also given as a fact that the user entered the correct password.

The statements must be represented using propositional logic and converted into Conjunctive Normal Form (CNF). The goal is to prove that the user is granted access. The negation of the goal is added to the knowledge base, and resolution is performed between clauses containing complementary literals. The intermediate result establishes authentication, which is then used to derive the final access-related conclusion.